



For Database Management Systems Final Project



WMATA

**(WASHINGTON METROPOLITAN
AREA TRANSIT AUTHORITY)**

CRIME ANALYSIS

KARY Metro Guards

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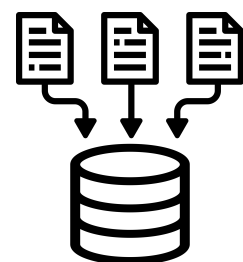
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Background



Data Source

Data Source:

- WMATA Performance Service Excellence Report
- WMATA Monthly Blotter Summaries (Crime Statistics)
 - Data from 2020 ~ 2022
- WMATA Metro Stations Dataset

Intended User



WMATA

The WMATA:

- Monitor crime, assaults, injuries, fatalities, and safety events by mode, station, and time of day.
- Identify high-risk stations, routes, and time periods and prioritize safety investments.
- Evaluate whether safety initiatives, policies, or operational changes are actually reducing incidents.

Intended User



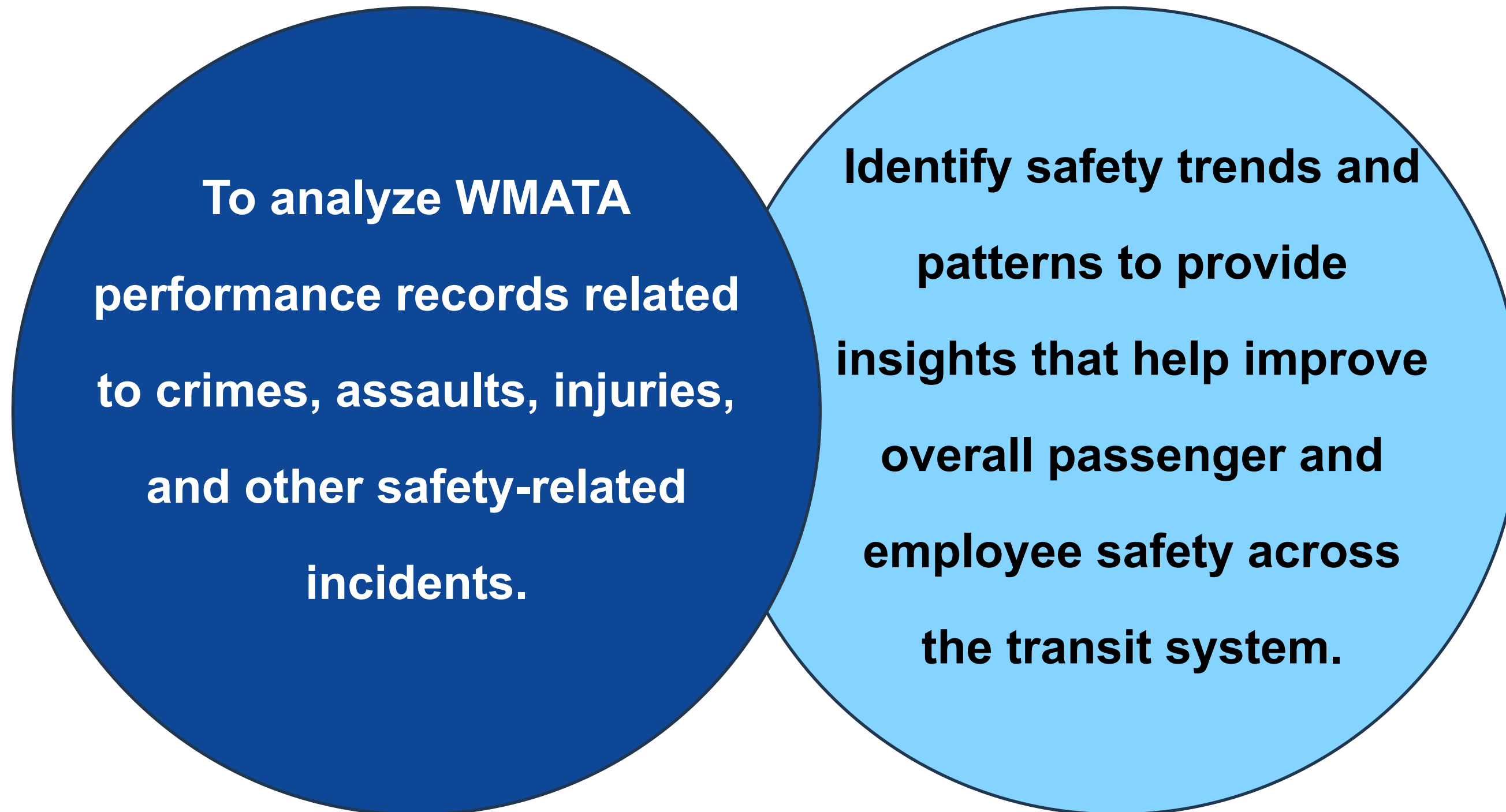
Law Enforcement

Metro Transit Police Department

- Understand which stations to prioritize patrols and police allocation.
- Monitor what type of crimes are more likely to occur in each location



Mission Statement





Mission Objective

To identify how incidents vary by time of day and between peak and off-peak periods.

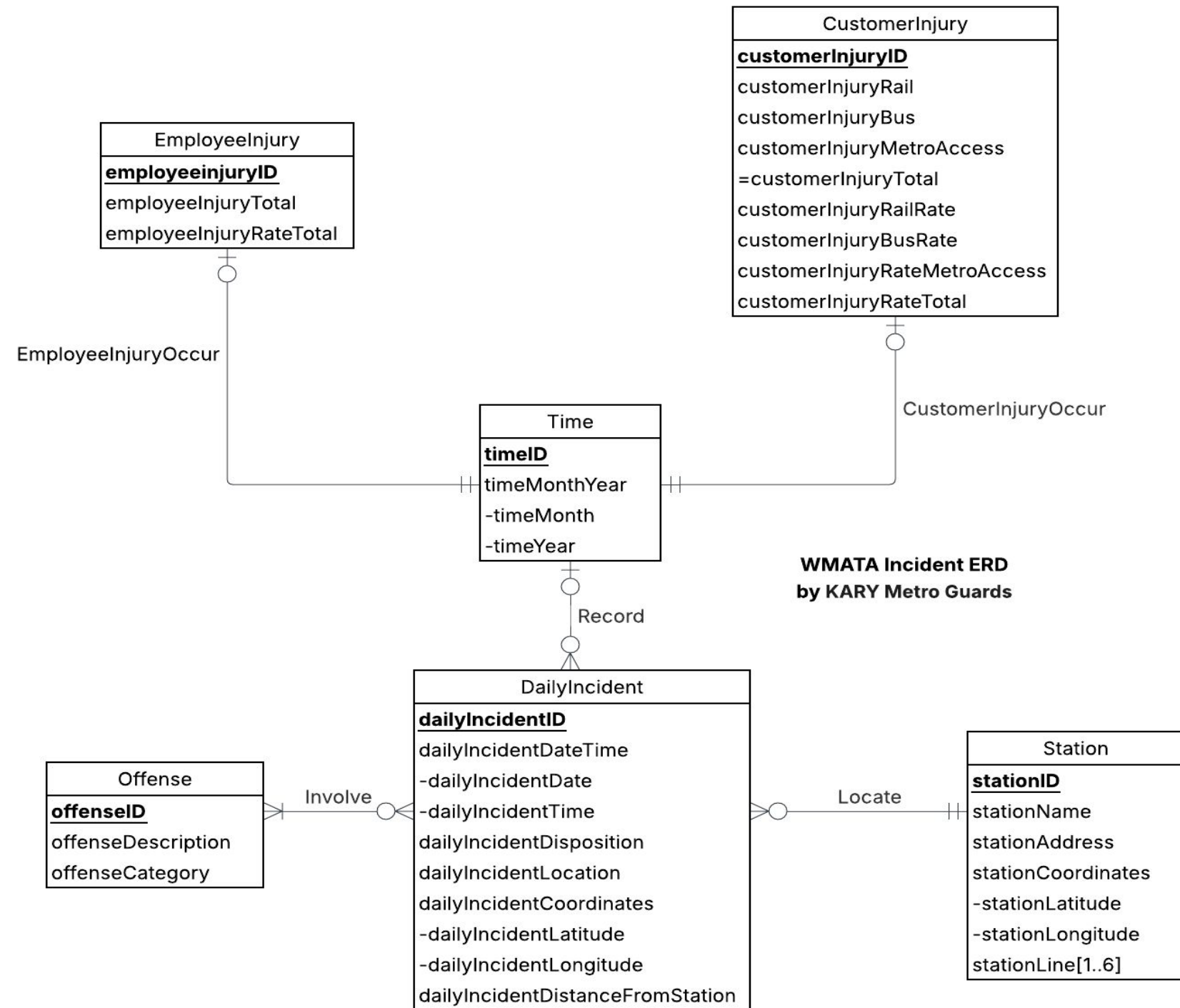
To examine how total crimes and injuries change over time from 2020 to 2022.

To determine which stations and rail lines have the highest frequency of daily incidents.

To identify the most frequent offense categories and how they change over time.

Conceptual Database

Entity Relationship Diagram (ERD)



Logical Database Design

Relational Schema

Time (**timeID**, timeYear, timeMonth)

CustomerInjury (**crimeID**, customerInjuryRail, customerInjuryBus, customerInjuryMetroAccess, customerInjuryRailRate, customerInjuryBusRate, customerInjuryMetroAccessRate, customerInjuryTotalRate, *timeID*)

EmployeeInjury (**employeeInjuryID**, employeeInjuryTotal, employeeInjuryTotalRate, *timeID*)

DailyIncident (**dailyIncidentID**, dailyIncidentDate, dailyIncidentTime, dailyIncidentDisposition, dailyIncidentLocation, dailyIncidentLatitude, dailyIncidentLongitude, dailyIncidentDistanceFromStation, *stationID*, *timeID*)

Offense (**offenseID**, offenseDescription, offenseCategory)

Involve (**dailyIncidentID**, **offenseID**)

Station (**stationID**, stationName, stationAddress, stationLatitude, stationLongitude)

StationLine (**stationID**, **stationLine**)

Physical Database Design

Time Table

```
CREATE TABLE Time (  
    timeID CHAR(7) NOT NULL,  
    timeYear INTEGER,  
    timeMonth VARCHAR(10),  
    CONSTRAINT pk_Time_timeID PRIMARY KEY (timeID) );
```

Employee Injury Table

```
CREATE TABLE EmployeeInjury (  
    employeeInjuryID CHAR(7) NOT NULL,  
    employeeInjuryTotal INTEGER,  
    employeeInjuryTotalRate DECIMAL(6,2),  
    timeID CHAR(7),  
    CONSTRAINT pk_EmployeeInjury_employeeInjuryID PRIMARY KEY (employeeInjuryID),  
    CONSTRAINT fk_EmployeeInjury_timeID FOREIGN KEY (timeID)  
        REFERENCES Time (timeID)  
        ON DELETE NO ACTION ON UPDATE CASCADE );
```


Physical Database Design

Customer Injury Table

```
CREATE TABLE CustomerInjury (  
    customerInjuryID CHAR(7) NOT NULL,  
    customerInjuryRail INTEGER,  
    customerInjuryBus INTEGER,  
    customerInjuryMetroAccess INTEGER,  
    customerInjuryRailRate DECIMAL(6,2),  
    customerInjuryBusRate DECIMAL(6,2),  
    customerInjuryMetroAccessRate DECIMAL(6,2),  
    customerInjuryTotalRate DECIMAL(6,2),  
    timeID CHAR(7),  
    CONSTRAINT pk_CustomerInjury_customerInjuryID PRIMARY KEY (customerInjuryID),  
    CONSTRAINT fk_CustomerInjury_timeID FOREIGN KEY (timeID)  
        REFERENCES Time (timeID)  
    ON DELETE NO ACTION ON UPDATE CASCADE );
```

Physical Database Design

Daily Incident Table

```
CREATE TABLE DailyIncident (  
    dailyIncidentID CHAR(8) NOT NULL,  
    dailyIncidentDate DATE,  
    dailyIncidentTime TIME,  
    dailyIncidentDisposition VARCHAR(50),  
    dailyIncidentLocation VARCHAR(100),  
    dailyIncidentLatitude DECIMAL(9,6),  
    dailyIncidentLongitude DECIMAL(9,6),  
    dailyIncidentDistanceFromStation DECIMAL(15,2),  
    stationID CHAR(4),  
    timeID CHAR(7),  
    CONSTRAINT pk_DailyIncident_dailyIncidentID PRIMARY KEY (dailyIncidentID),  
    CONSTRAINT fk_DailyIncident_stationID FOREIGN KEY (stationID)  
        REFERENCES Station (stationID)  
        ON DELETE NO ACTION ON UPDATE CASCADE,  
    CONSTRAINT fk_DailyIncident_timeID FOREIGN KEY (timeID)  
        REFERENCES Time (timeID)  
        ON DELETE NO ACTION ON UPDATE CASCADE );
```

Physical Database Design

Station Table

```
CREATE TABLE Station (  
    stationID CHAR(4) NOT NULL,  
    stationName VARCHAR(50),  
    stationAddress VARCHAR(100),  
    stationLatitude DECIMAL(9,6),  
    stationLongitude DECIMAL(9,6),  
    CONSTRAINT pk_Station_stationID PRIMARY KEY (stationID));
```

Station Line Table

```
CREATE TABLE StationLine (  
    stationID CHAR(4) NOT NULL,  
    stationLine VARCHAR(10),  
    CONSTRAINT pk_StationLine_stationID_stationLine PRIMARY KEY (stationID, stationLine),  
    CONSTRAINT fk_StationLine_stationID FOREIGN KEY (stationID)  
        REFERENCES Station (stationID)  
    ON DELETE NO ACTION ON UPDATE CASCADE );
```

Physical Database Design

Offense Table

```
CREATE TABLE Offense (  
    offenseID CHAR(6) NOT NULL,  
    offenseDescription VARCHAR(150),  
    CONSTRAINT pk_Offense_offenseID PRIMARY KEY (offenseID) );
```

Involve Table

```
CREATE TABLE Involve (  
    dailyIncidentID CHAR(8) NOT NULL,  
    offenseID CHAR(6) NOT NULL,  
    CONSTRAINT pk_Involve_dailyIncidentID_offenseID PRIMARY KEY (dailyIncidentID, offenseID),  
    CONSTRAINT fk_Involve_dailyIncidentID FOREIGN KEY (dailyIncidentID)  
        REFERENCES DailyIncident (dailyIncidentID)  
        ON DELETE NO ACTION ON UPDATE CASCADE,  
    CONSTRAINT fk_Involve_offenseID FOREIGN KEY (offenseID)  
        REFERENCES Offense (offenseID)  
        ON DELETE NO ACTION ON UPDATE CASCADE );
```

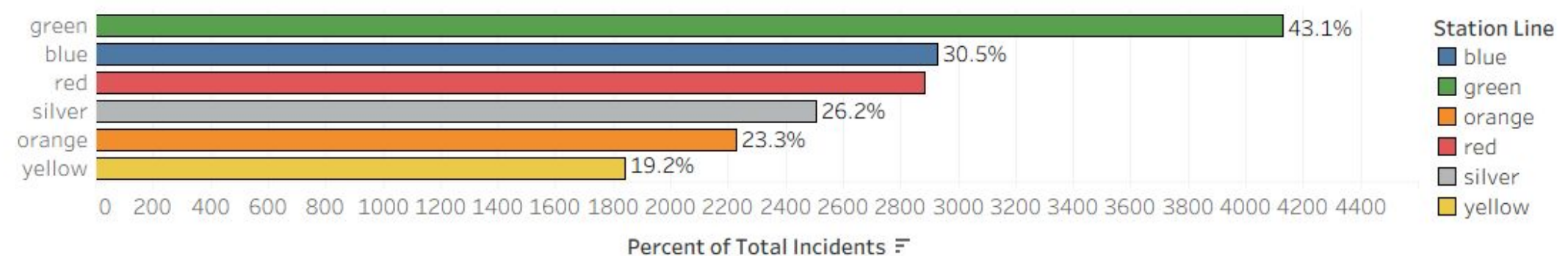
Business Transaction 1

What is the share of total incidents across all station lines?

```
SELECT sl.stationLine AS 'Station Line',  
       CAST(ROUND(COUNT(d.dailyIncidentID) * 100.0 /  
       (SELECT COUNT(*) FROM DailyIncidentNearStation),1) AS FLOAT)  
       AS 'Percent of Total Incidents'  
FROM DailyIncidentNearStation d  
     JOIN station s ON s.stationID = d.stationID  
     JOIN StationLine sl ON sl.stationID = s.stationID  
GROUP BY sl.stationLine  
ORDER BY [Percent of Total Incidents] DESC;
```

	Station Line	Percent of Total Incidents
1	green	43.1
2	blue	30.5
3	red	30.1
4	silver	26.2
5	orange	23.3
6	yellow	19.2

Share of Total Incidents across Station Lines



Business Transaction 2

What are the top stations by number of incidents and how do they differ during different time periods?

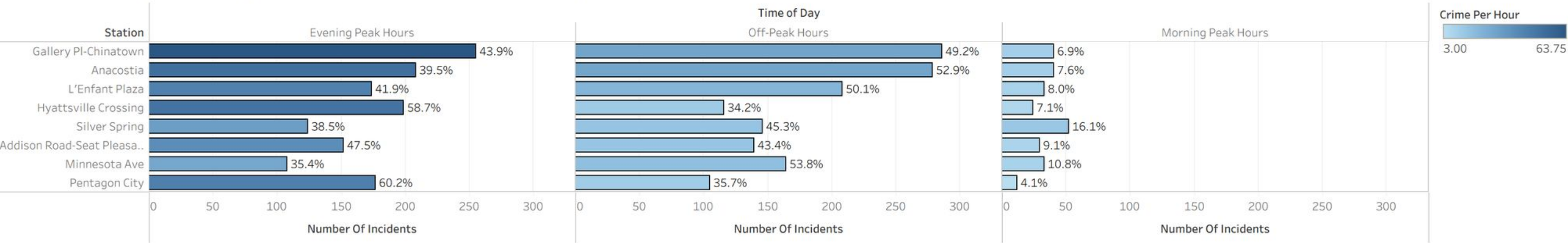
```
WITH station_incident_time_classification AS (  
    SELECT s.stationName, d.dailyIncidentID,  
    CASE  
        WHEN DATEPART(hour, dailyIncidentTime) BETWEEN 6 AND 9 THEN 'Morning Peak Hours'  
        WHEN DATEPART(hour, dailyIncidentTime) BETWEEN 15 AND 18 THEN 'Evening Peak Hours'  
        ELSE 'Off-Peak Hours'  
    END AS time_classification  
    FROM DailyIncidentNearStation d  
    JOIN station s ON d.stationID = s.stationID ),  
  
station_total_incidents AS (  
    SELECT stationName, COUNT(dailyIncidentID) AS total_incidents  
    FROM station_incident_time_classification  
    GROUP BY stationName )  
  
SELECT sc.stationName AS 'Station', sc.time_classification AS 'Time of Day',  
    CAST(ROUND(COUNT(sc.dailyIncidentID) * 100.0 / sti.total_incidents, 1) AS FLOAT) AS  
    'Percent of Total Incidents',  
    CASE  
        WHEN sc.time_classification = 'Off-Peak Hours' THEN CAST(COUNT(sc.dailyIncidentID) / 10.0 AS FLOAT)  
        WHEN sc.time_classification = 'Morning Peak Hours' THEN CAST(COUNT(sc.dailyIncidentID) / 4.0 AS FLOAT)  
        WHEN sc.time_classification = 'Evening Peak Hours' THEN CAST(COUNT(sc.dailyIncidentID) / 4.0 AS FLOAT)  
    END AS 'Number of Crimes per Hour'  
FROM station_incident_time_classification sc  
    JOIN station_total_incidents sti ON sc.stationName = sti.stationName  
GROUP BY sc.stationName, sc.time_classification, sti.total_incidents  
ORDER BY sti.total_incidents DESC, [Number of Crimes per Hour] DESC;
```


Business Transaction 2

What are the top stations by number of incidents and how do they differ during different time periods?

	Station	Time of Day	Percent of Total Incidents	Number of Crimes per Hour
1	Gallery Pl-Chinatown	Evening Peak Hours	43.9	63.75
2	Gallery Pl-Chinatown	Off-Peak Hours	49.2	28.6
3	Gallery Pl-Chinatown	Morning Peak Hours	6.9	10
4	Anacostia	Evening Peak Hours	39.5	52
5	Anacostia	Off-Peak Hours	52.9	27.9
6	Anacostia	Morning Peak Hours	7.6	10
7	L'Enfant Plaza	Evening Peak Hours	41.9	43.5
8	L'Enfant Plaza	Off-Peak Hours	50.1	20.8
9	L'Enfant Plaza	Morning Peak Hours	8	8.25
10	Hyattsville Crossing	Evening Peak Hours	58.7	49.75
11	Hyattsville Crossing	Off-Peak Hours	34.2	11.6
12	Hyattsville Crossing	Morning Peak Hours	7.1	6
13	Silver Spring	Evening Peak Hours	38.5	31
14	Silver Spring	Off-Peak Hours	45.3	14.6
15	Silver Spring	Morning Peak Hours	16.1	13
16	Addison Road-Sea...	Evening Peak Hours	47.5	38
17	Addison Road-Sea...	Off-Peak Hours	43.4	13.9
18	Addison Road-Sea...	Morning Peak Hours	9.1	7.25
19	Minnesota Ave	Evening Peak Hours	35.4	27
20	Minnesota Ave	Off-Peak Hours	53.8	16.4
21	Minnesota Ave	Morning Peak Hours	10.8	8.25
22	Pentagon City	Evening Peak Hours	60.2	44.25
23	Pentagon City	Off-Peak Hours	35.7	10.5
24	Pentagon City	Morning Peak Hours	4.1	3

What are the top stations by number of incidents and how do they differ during different time periods?



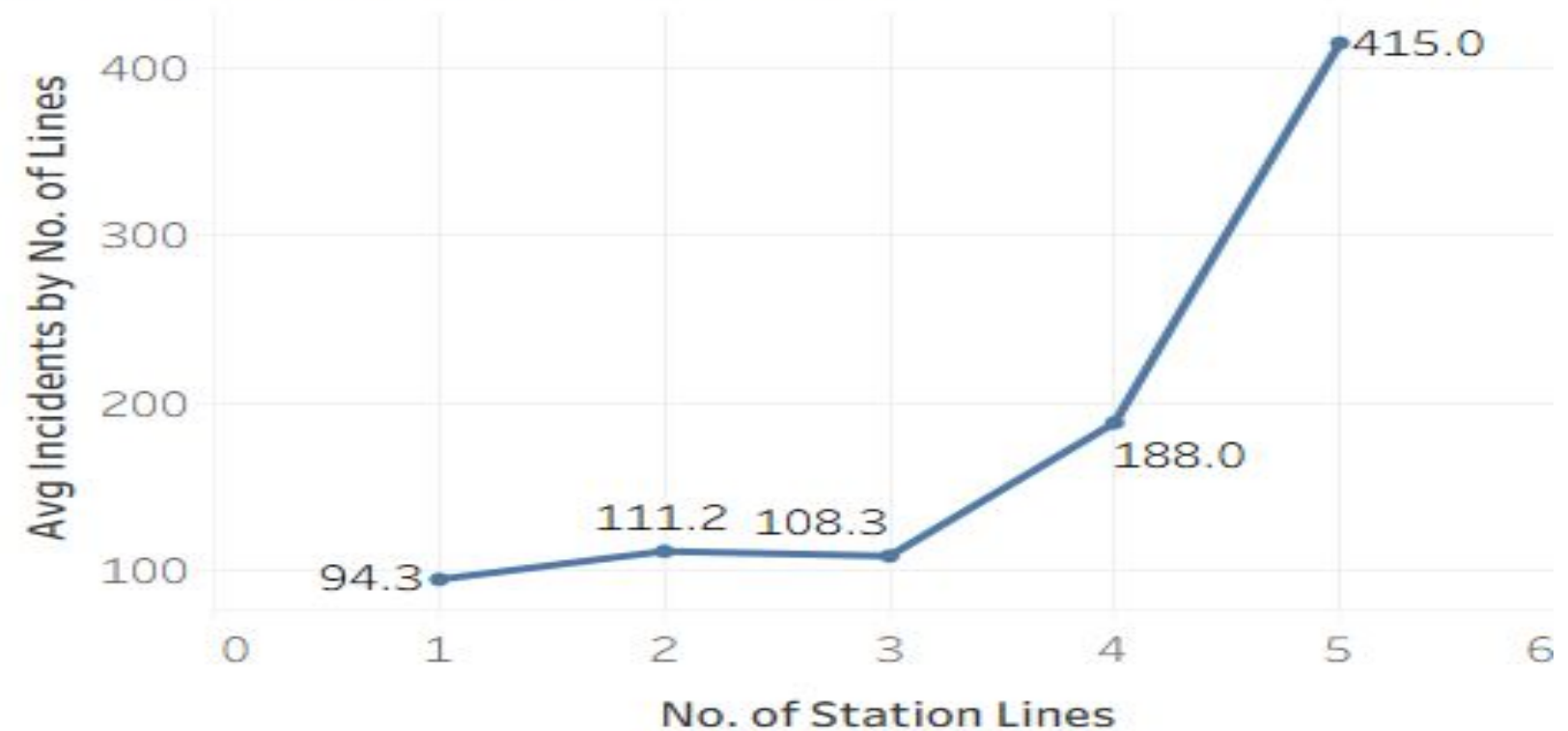
Business Transaction 3

Do stations with more lines tend to have more incidents?

```
WITH lines AS (  
    SELECT s.stationID, s.stationName,  
           COUNT(sl.stationLine) AS number_of_station_lines  
    FROM Station s  
    JOIN StationLine sl ON s.stationID = sl.stationID  
    GROUP BY s.stationID, s.stationName ),  
  
incidents AS (  
    SELECT s.stationID, s.stationName,  
           COUNT(d.dailyIncidentID) AS number_of_incidents  
    FROM DailyIncidentNearStation d  
    JOIN Station s ON s.stationID = d.stationID  
    GROUP BY s.stationID, s.stationName )  
  
SELECT l.number_of_station_lines AS 'No. of Station Lines',  
       CAST(ROUND(AVG(i.number_of_incidents * 1.0), 1) AS FLOAT)  
AS 'Avg Incidents per No. of Lines'  
FROM lines l  
     JOIN incidents i ON l.stationID = i.stationID  
GROUP BY l.number_of_station_lines  
ORDER BY l.number_of_station_lines;
```

	No. of Station Lines	Avg Incidents per No. of Lines
1	1	94.3
2	2	111.2
3	3	108.3
4	4	188
5	5	415

Avg No. of Incidents per No. of Lines



Business Transaction 4

What are the top 10 most frequently occurring offense categories?

```
SELECT TOP 10 f.offenseCategory AS 'Offense Type',  
             COUNT(d.dailyIncidentID) AS 'Total Number of Incidents'  
FROM DailyIncidentNearStation d  
     JOIN Involve i ON i.dailyIncidentID = d.dailyIncidentID  
     JOIN Offense f ON f.offenseID = i.offenseID  
GROUP BY f.offenseCategory  
ORDER BY [Total Number of Incidents] DESC
```

	Offense Type	Total Number of Incidents
1	Violent Crimes	2137
2	Fare Evasion	1937
3	Theft / Property Crimes	1463
4	Vandalism / Property Damage	962
5	Justice / Legal Obstruction	787
6	Alcohol Offenses	779
7	Sexual Offenses	600
8	Disorderly / Public Disturbance	591
9	Drug Offenses	514
10	Burglary / Trespass	494

Most common offense types

Violent Crimes	Theft / Property Crimes	Alcohol Offenses	Justice / Legal	Sexual
Fare Evasion	Vandalism / Property Damage	Disorderly / Public Disturbance	Drug Offenses	
		Burglary / Trespass		

Business Transaction 5, 6 (KPIs for Dashboard 1)

Business Transaction 5: What are the total number of incidents?

```
SELECT COUNT(dailyIncidentID) AS 'Total Incidents'
FROM DailyIncidentNearStation
```

	Total Incidents
1	9581

Total Incidents
9,581

Business Transaction 6: What is the share of total incidents by time of the day?

```
WITH IncidentTimeOfDay AS (
  SELECT d.dailyIncidentID,
    CASE
      WHEN DATEPART(hour, dailyIncidentTime) BETWEEN 6 AND 9 THEN 'Morning Peak Hours'
      WHEN DATEPART(hour, dailyIncidentTime) BETWEEN 15 AND 18 THEN 'Evening Peak Hours'
      ELSE 'Off-Peak Hours'
    END AS time_of_day
  FROM DailyIncidentNearStation d )
```

	Time of Day	Share of Total Incidents
1	Evening Peak Hours	40.25
2	Morning Peak Hours	11.72
3	Off-Peak Hours	48.03

```
SELECT i.time_of_day AS 'Time of Day',
  CAST(ROUND(COUNT(i.dailyIncidentID) * 100.0 /
    (SELECT COUNT(dailyIncidentID) FROM IncidentTimeOfDay), 2) AS FLOAT) AS 'Share of Total Incidents'
FROM IncidentTimeOfDay i
GROUP BY i.time_of_day
```

Share of Total Incidents		
Evening Peak	Morning Peak	Off-Peak
40.25%	11.72%	48.03%

Business Transaction 7, 8 (KPIs for Dashboard 1)

Business Transaction 7: What is the station with the most number of incidents?

```
SELECT TOP 1 s.stationName AS 'Station',  
COUNT(d.dailyIncidentID) 'Total No. of Incidents'  
FROM DailyIncidentNearStation d  
    JOIN Station s ON s.stationID = d.stationID  
GROUP BY s.stationName  
ORDER BY [Total No. of Incidents] DESC
```

	Station	Total No. of Incidents
1	Gallery PI-Chinatown	581

Top Station	
Gallery PI-Chinatown	581

Business Transaction 8: What station line has the most number of incidents?

```
SELECT TOP 1 sl.stationLine AS 'Station Line',  
COUNT(d.dailyIncidentID) AS 'Number of Incidents'  
FROM DailyIncidentNearStation d  
    JOIN StationLine sl ON sl.stationID = d.stationID  
GROUP BY sl.stationLine  
ORDER BY [Number of Incidents] DESC
```

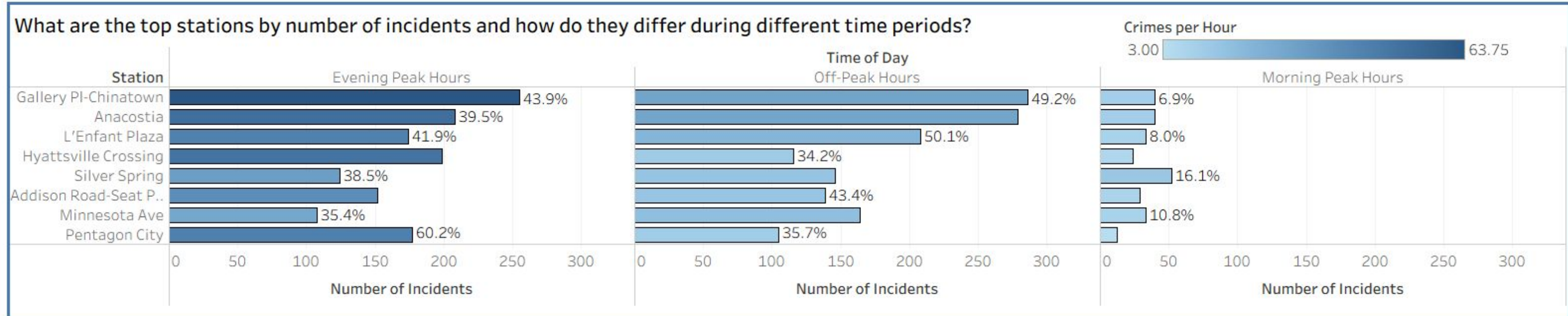
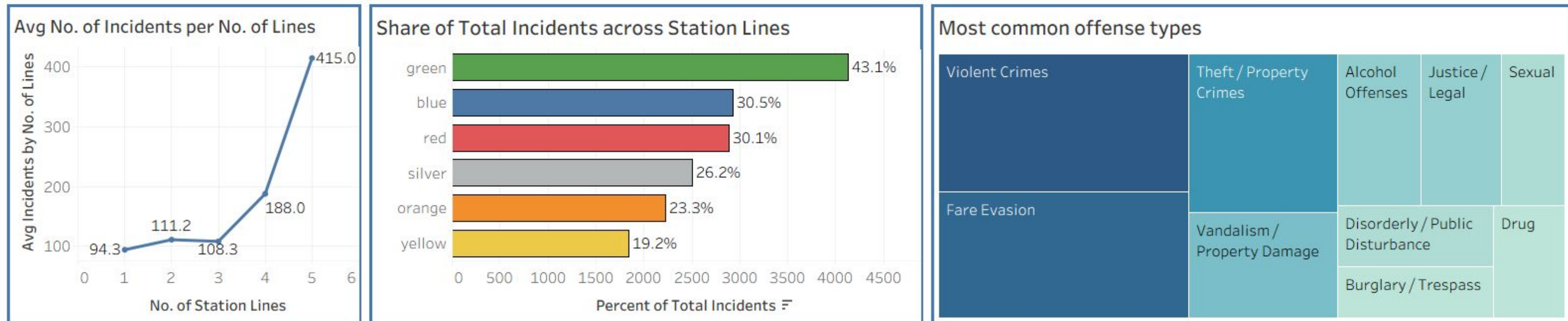
	Station Line	Number of Incidents
1	green	4130

Top Station Line	
green	4,130

Dashboard 1

WMATA Station Incident Dashboard | 2020 - 2022

Total Incidents	Share of Total Incidents			Top Station		Top Station Line	
9,581	Evening Peak 40.25%	Morning Peak 11.72%	Off-Peak 48.03%	Gallery PI-Chinatown	581	green	4,130



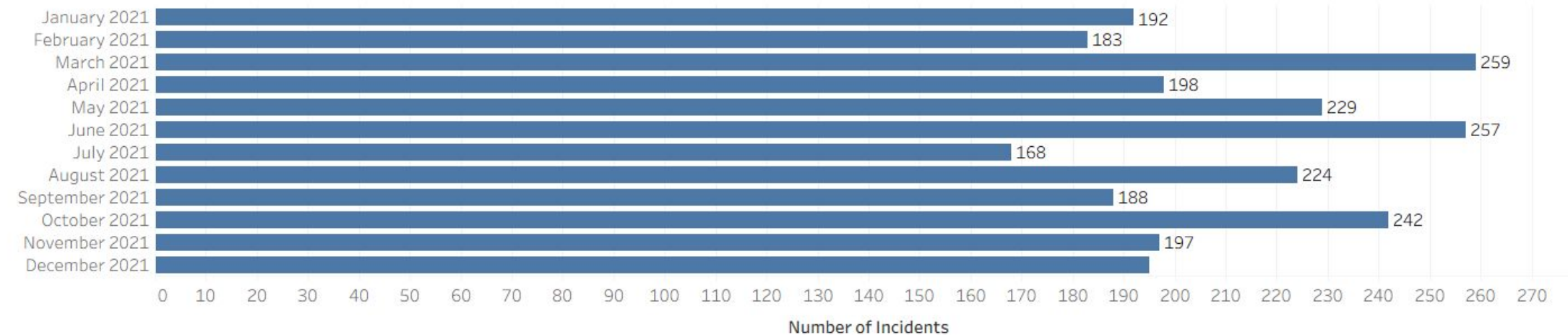
Business Transaction 9

What is the total number of crime over the months?

```
SELECT t.timeYear AS 'Year' ,t.timeMonth AS 'Month', COUNT(d.dailyIncidentID) AS 'Number of Incidents'
FROM DailyIncidentNearStation d, [Time] t
WHERE t.timeID = d.timeID
GROUP BY t.timeYear,t.timeMonth, MONTH(d.dailyIncidentDate)
ORDER BY t.timeYear, MONTH(d.dailyIncidentDate), COUNT(d.dailyIncidentID) DESC
```

	Year	Month	Number of Incidents
1	2020	January	903
2	2020	February	797
3	2020	March	398
4	2020	April	125
5	2020	May	157
6	2020	June	199
7	2020	July	168
8	2020	August	194
9	2020	September	185
10	2020	October	314
11	2020	November	302
12	2020	December	261
13	2021	January	192
14	2021	February	183

What is the total number of crime over the months?



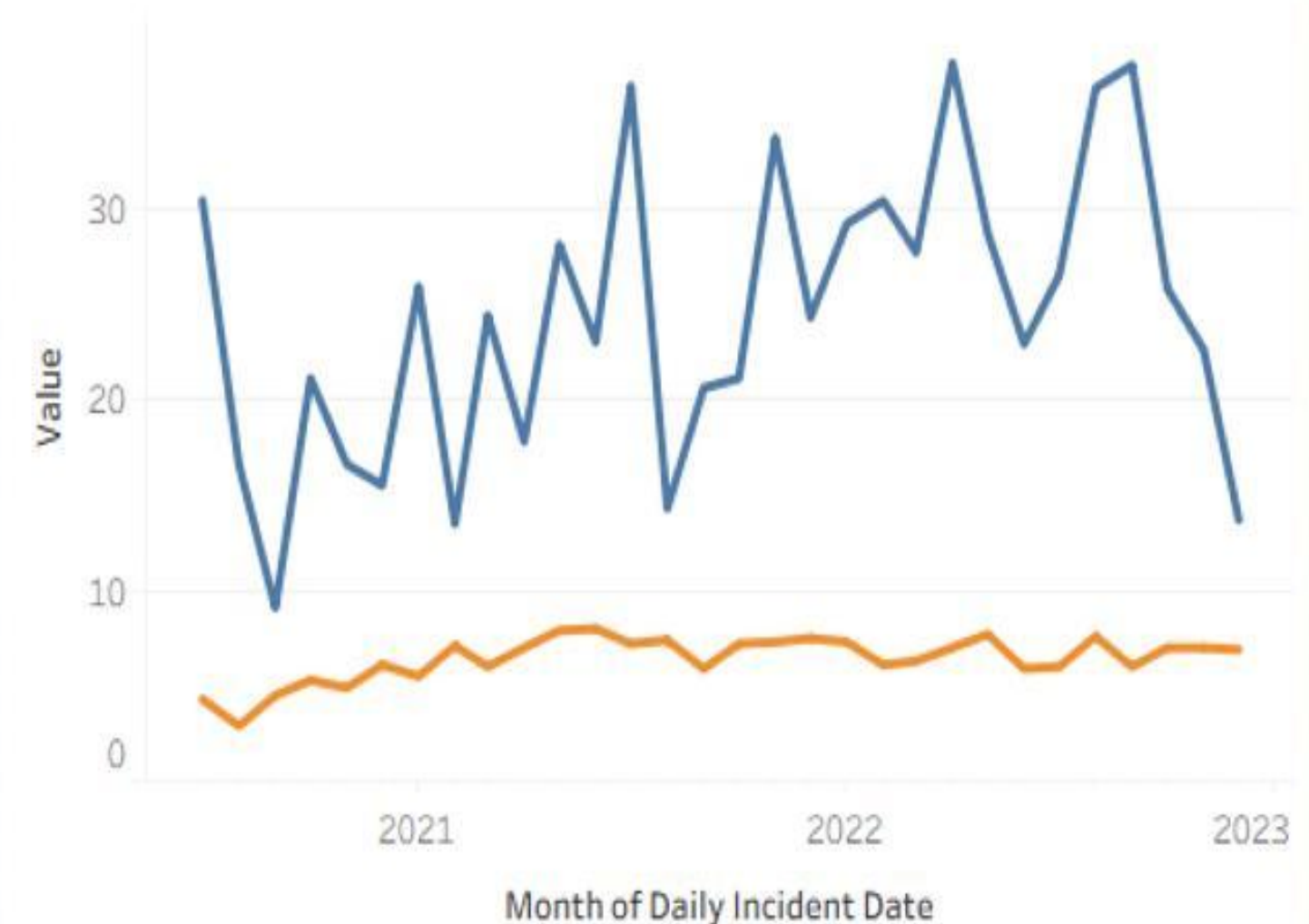
Business Transaction 10

How has customer and employee injury changed over time?

```
SELECT t.timeYear AS 'Year', t.timeMonth AS 'Month',  
       ci.customerInjuryTotalRate AS 'Total Customer Injury Rate',  
       ei.employeeInjuryTotalRate AS 'Total Employee Injury Rate'  
FROM Time t  
      JOIN CustomerInjury ci ON ci.timeID = t.timeID  
      JOIN EmployeeInjury ei ON ei.timeID = t.timeID  
ORDER BY t.timeYear;
```

	Year	Month	Total Customer Injury Rate	Total Employee Injury Rate
1	2020	July	30.40	4.30
2	2020	August	16.60	2.90
3	2020	September	9.10	4.50
4	2020	October	21.10	5.30
5	2020	November	16.60	4.90
6	2020	December	15.50	6.10
7	2021	January	25.90	5.50
8	2021	Febuary	13.50	7.10
9	2021	March	24.40	6.00
10	2021	April	17.80	7.00
11	2021	May	28.10	7.90
12	2021	June	23.00	8.00

How has customer and employee injury change over time?



Business Transaction 11

How have the most frequent types of offenses changed over the months?

```
WITH CrimeByMonth AS (  
    SELECT YEAR(d.dailyIncidentDate) AS dailyIncidentYear,  
           MONTH(d.dailyIncidentDate) AS dailyIncidentMonthNumber,  
           DATENAME(MONTH, d.dailyIncidentDate) AS dailyIncidentMonth,  
           o.offenseCategory AS offenseType,  
           COUNT(d.dailyIncidentID) AS offenseCount  
    FROM DailyIncident d  
    JOIN Involve i ON d.dailyIncidentID = i.dailyIncidentID  
    JOIN Offense o ON i.offenseID = o.offenseID  
    GROUP BY YEAR(d.dailyIncidentDate), MONTH(d.dailyIncidentDate),  
             DATENAME(MONTH, d.dailyIncidentDate), o.offenseCategory ),
```

```
YearlyCrimeTotals AS (  
    SELECT dailyIncidentYear, offenseType,  
           SUM(offenseCount) AS totalYearlyOffense  
    FROM CrimeByMonth  
    GROUP BY dailyIncidentYear, offenseType ),
```

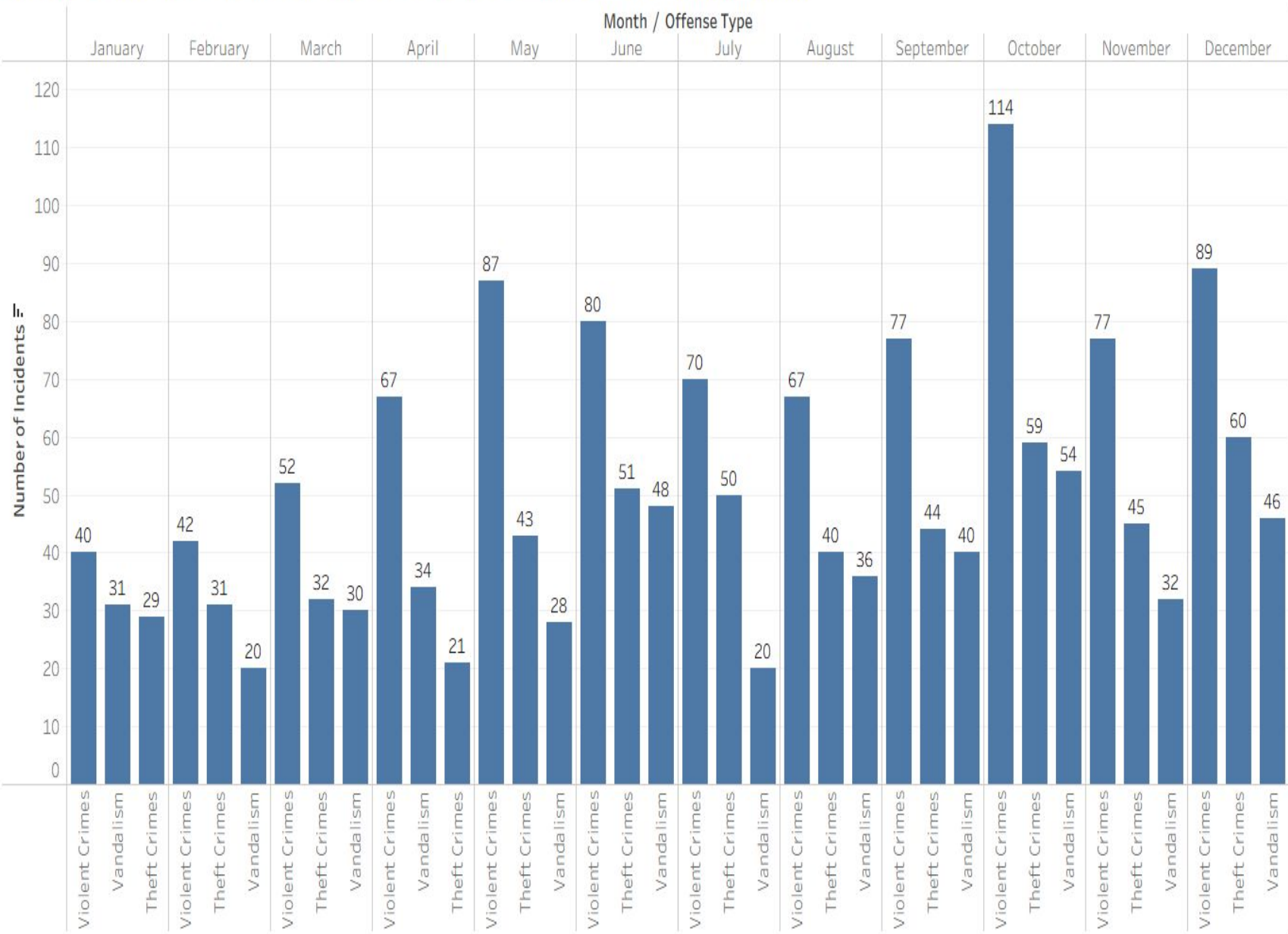
```
RankedCrimes AS (  
    SELECT dailyIncidentYear, offenseType, totalYearlyOffense,  
           DENSE_RANK() OVER (PARTITION BY dailyIncidentYear  
                               ORDER BY totalYearlyOffense DESC) AS CrimeRank  
    FROM YearlyCrimeTotals )
```

```
SELECT cbm.dailyIncidentYear AS "Year",  
       cbm.dailyIncidentMonthNumber AS "Month Number",  
       cbm.dailyIncidentMonth AS "Month",  
       cbm.offenseType AS "Offense Type",  
       cbm.offenseCount AS "Number of Incidents"  
FROM CrimeByMonth cbm  
     JOIN RankedCrimes rc ON cbm.dailyIncidentYear = rc.dailyIncidentYear  
     AND cbm.offenseType = rc.offenseType  
WHERE rc.CrimeRank <= 3  
ORDER BY cbm.dailyIncidentYear, cbm.dailyIncidentMonthNumber, cbm.offenseCount  
DESC;
```


Business Transaction 11

How have the most frequent types of offenses changed over the months?

How have the most frequent types of offenses changed over the months?



	Year	Month Number	Month	Offense Type	Number of Incidents
1	2020	1	January	Fare Evasion	561
2	2020	1	January	Alcohol Offenses	139
3	2020	1	January	Violent Crimes	96
4	2020	2	February	Fare Evasion	450
5	2020	2	February	Alcohol Offenses	141
6	2020	2	February	Violent Crimes	95
7	2020	3	March	Fare Evasion	177
8	2020	3	March	Alcohol Offenses	71
9	2020	3	March	Violent Crimes	65
10	2020	4	April	Violent Crimes	31
11	2020	4	April	Alcohol Offenses	7
12	2020	4	April	Fare Evasion	6
13	2020	5	May	Violent Crimes	38
14	2020	5	May	Alcohol Offenses	25
15	2020	6	June	Violent Crimes	48

Dashboard 2

WMATA Incident and Injury Dashboard | 2020 - 2022

Total Incidents

2,532

Most Repeated
Crime

Violent Crimes

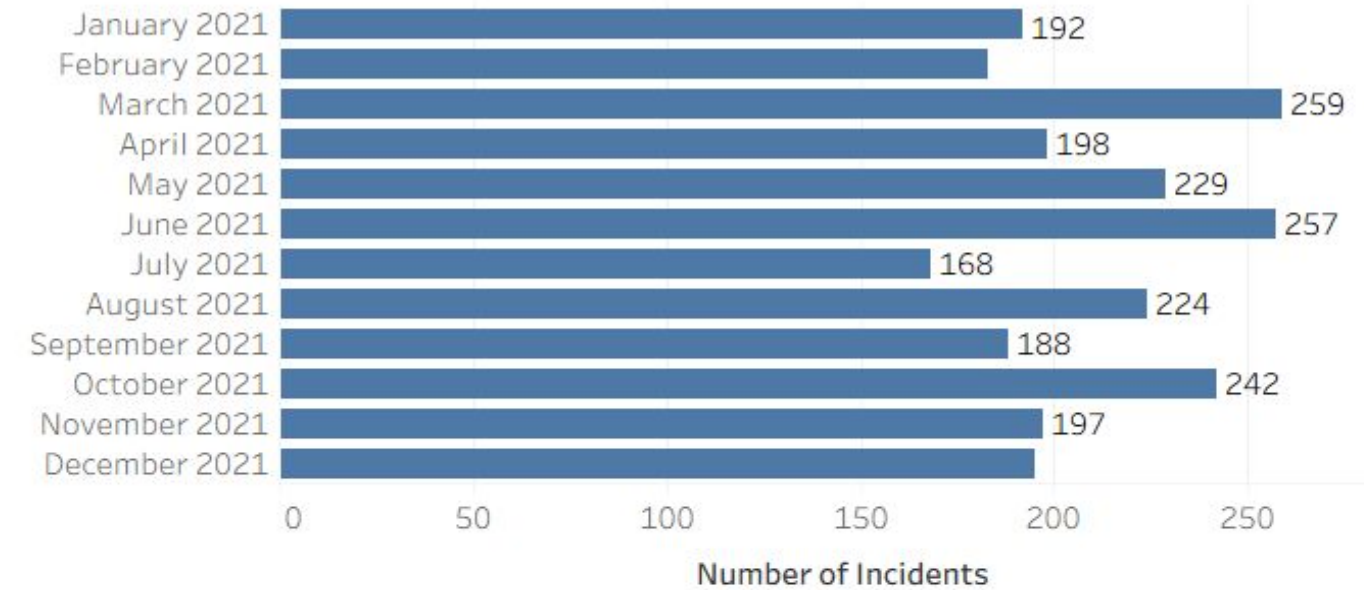
Measure Names

Customer Injury Total Rate
Employee Injury Total Rate

How has customer and employee injury changed over time?



What is the total number of crime over the months?

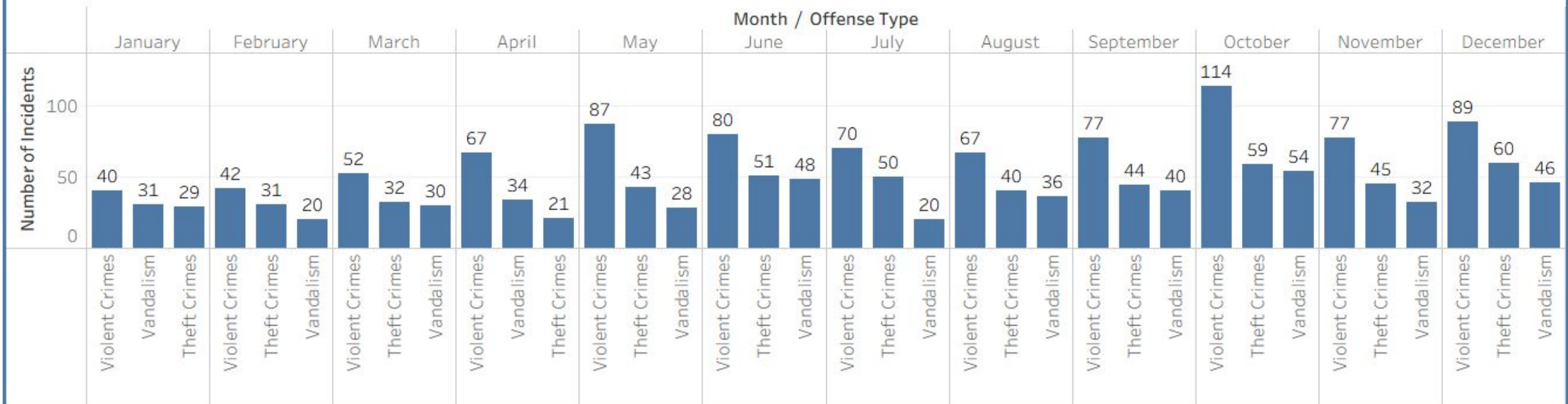


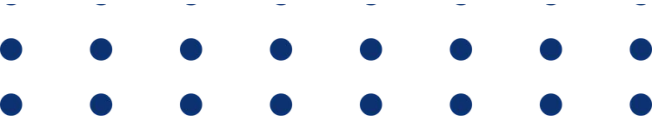
2020

2021

2022

How have the most frequent types of offenses changed over the months?





Insights



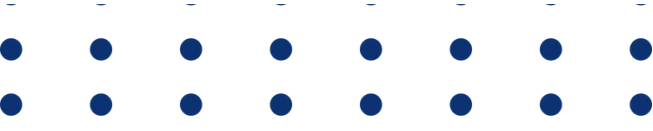
Q1. How do incidents vary by time of day and between peak and off-peak periods?

- **Only about 11.75% of incidents occur during morning peak (6–9AM), while roughly 40.3% occur in evening peak and 48.0% in off-peak hours.**
- **As evening peak is a shorter window, it has the highest incident rate per hour, making late afternoon and early evening the riskiest time to ride.**



Q2. How do total crimes and injuries change over time from 2020 to 2022?

- **From 2020–2022 WMATA recorded 9,581 crime incidents, with a sharp drop in early 2020 and a rebound in 2022 as COVID restrictions eased and ridership returned.**
- **Customer injuries are consistently higher and more volatile than employee injuries and trend upward into 2022, so overall safety conditions are driven mainly by crime and customer injury patterns.**



Insights



Q3. Which stations and rail lines have the highest frequency of daily incidents?

- **The Green Line is the highest-risk line, responsible for about 43% of all reported incidents and 4,130 incidents overall.**
- **Multi-line hubs such as Gallery Place–Chinatown show the highest station totals, with incident counts rising sharply at stations served by four or five lines, making these the primary focus for safety efforts.**



Q4. What are the most frequent offense categories and how do they change over time?

- **Violent Crimes, Fare Evasion, and Theft/Property Crimes are the dominant categories, with Vandalism/Property Damage, Justice/Legal Obstruction, and Alcohol Offenses forming a second tier.**
- **Offense patterns shift over time: Fare Evasion dominates early 2020, all major categories drop with COVID-related ridership declines, then Violent Crimes and Alcohol Offenses become more prominent again as ridership recovers.**

Insights

Recommendations

- **Prioritize high-risk lines and stations**
- **Focus safety resources on the Green Line and major transfer hubs like Gallery Place–Chinatown, L’Enfant Plaza, and other multi-line stations, which account for a disproportionate share of incidents.**
- **Conduct deeper analysis on why the Green Line has higher incident levels (surrounding area crime, station design, crowding, etc.) and tailor interventions accordingly.**
- **Shift some resources away from lower-incident lines (e.g., Yellow Line) and toward Green Line and key hubs, rather than staffing all lines uniformly.**
- **Align staffing and enforcement with time-of-day risk (BT2, BT6)**
- **Increase Metro Transit Police presence, station staff, and supervisors during evening peak and key off-peak periods, instead of treating morning peak as the primary focus.**
- **Use dynamic staffing schedules so that more officers and staff are on duty from roughly 3PM–7PM and in late off-peak hours at high-incident stations.**
- **Enhance CCTV monitoring, visibility of cameras, and overhead lighting during evening and late off-peak hours, especially at Green Line and transfer stations.**

Insights

»»» Recommendations

- Target the main offense types instead of treating all incidents equally
- Design specific strategies for Violent Crimes, Fare Evasion, and Theft/Property Crimes, since they drive much of the safety burden (e.g., focused fare enforcement, theft deterrence campaigns, visible patrols).
- Use monthly offense trends to time short, focused enforcement or outreach surges in periods when particular offenses historically spike (e.g., seasonal fare evasion or alcohol-related patterns).
- Improve how WMATA measures and evaluates safety performance
- Incorporate ridership and staffing data so customer and employee injuries can be expressed as rates, not just counts (incidents per million riders, incidents per employee hour).
- Use these normalized metrics to evaluate whether safety initiatives are actually reducing risk, rather than just reflecting changes in ridership or exposure.
- Continue tracking crime and injury trends by time of day, location, and offense category to support iterative, data-driven adjustments to safety strategy.

THANK YOU



Feel Free to Ask Us Questions!